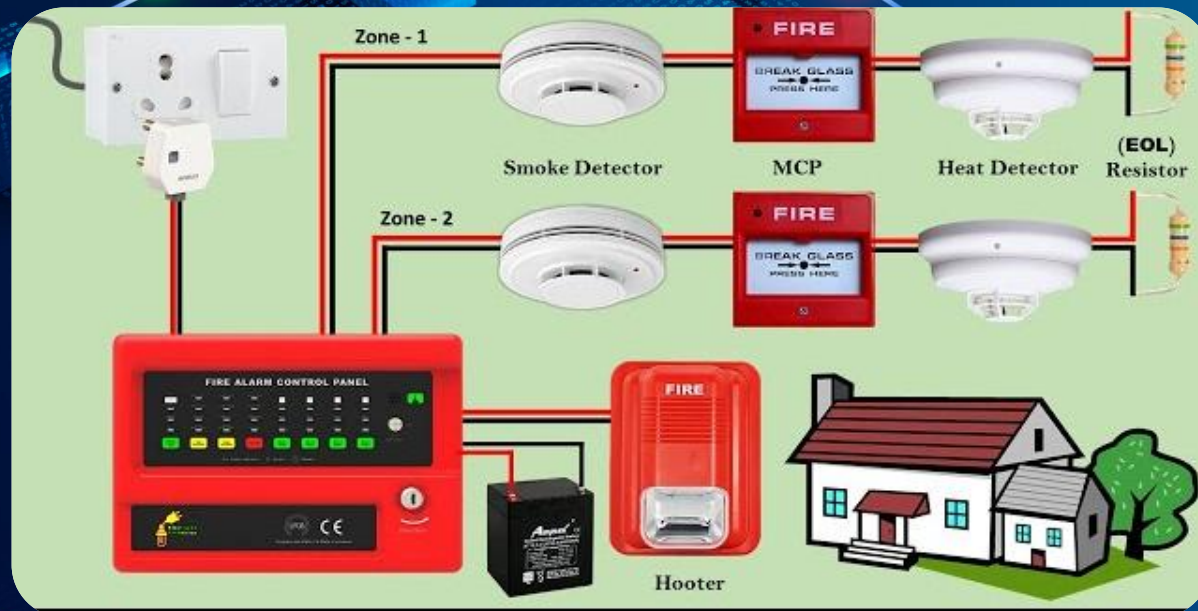


SMART FIRE / SMOKE WARNING SYSTEM



V Priyangan ID: 25197298

PROJECT OVERVIEW



- **Title:** Smart Fire / Smoke Warning System
- **Subtitle:** An IoT-Based Smart Safety Solution

- **Student Name:** Priyangan Vasantharajan
- **Student ID:** 25197298
- **Course/Module:** Computer Science With AI

- **Date:** 16 March 2026

- **Smart Product Category:** Smart Home Safety & Automation
- **UN SDGs:**
 - ● SDG 3 — Good Health & Well-being
 - ● SDG 11 — Sustainable Cities & Communities

PROBLEM & CONTEXT

"Every 6 seconds, a fire department responds to a fire somewhere in the world. Many of these incidents could be prevented or minimised with smarter detection systems."

- Traditional smoke alarms are **reactive, not smart** — they only beep locally
- They cannot **send remote alerts**, log events, or integrate with other systems
- Many fire fatalities occur at **night or when residents are away** — when no one hears the alarm
- A smart system solves this by combining **real-time sensing, automated responses, and connectivity**
- **Why it matters:** Protects lives, reduces property damage, and supports safer communities — directly aligning with SDG 3 and SDG 11

Research Summary

	Source	Key Finding	Trend
1	Kumar et al. (2021) – IEEE	IoT-based fire systems reduce response time by up to 40%	IoT
2	Figaro Engineering (2020)	MQ-2 sensor detects LPG, smoke, CO with high sensitivity	Edge Sensing
3	Monk, S. (2022) – <i>Programming Arduino</i>	Arduino is ideal for rapid embedded safety prototypes	Embedded Systems
4	Home Office, UK Gov (2023)	Over 37,000 accidental dwelling fires reported in England alone	Real-world Need

- Research confirms that IoT + edge computing enables faster, localised fire detection without cloud dependency
- Validates the use of Arduino + MQ-2 as a reliable, low-cost smart solution

Product Functionality

What the product does:

A smart alarm system that continuously monitors air quality and temperature, triggering immediate audio and visual alerts when dangerous levels are detected.

Inputs:

- MQ-2 Smoke Sensor – detects smoke and gas particles in the air
- LM35 / DHT11 Temperature Sensor – monitors ambient heat levels

Processing:

Arduino Uno reads sensor values and compares against safety thresholds
If smoke OR temperature exceeds limit → alarm sequence is triggered

Outputs:

- Buzzer – loud audio alert
- Red LED – visual warning indicator
- LCD Display (*optional*) – shows "FIRE DETECTED" message

SYSTEM IDEA

INPUT SENSORS:

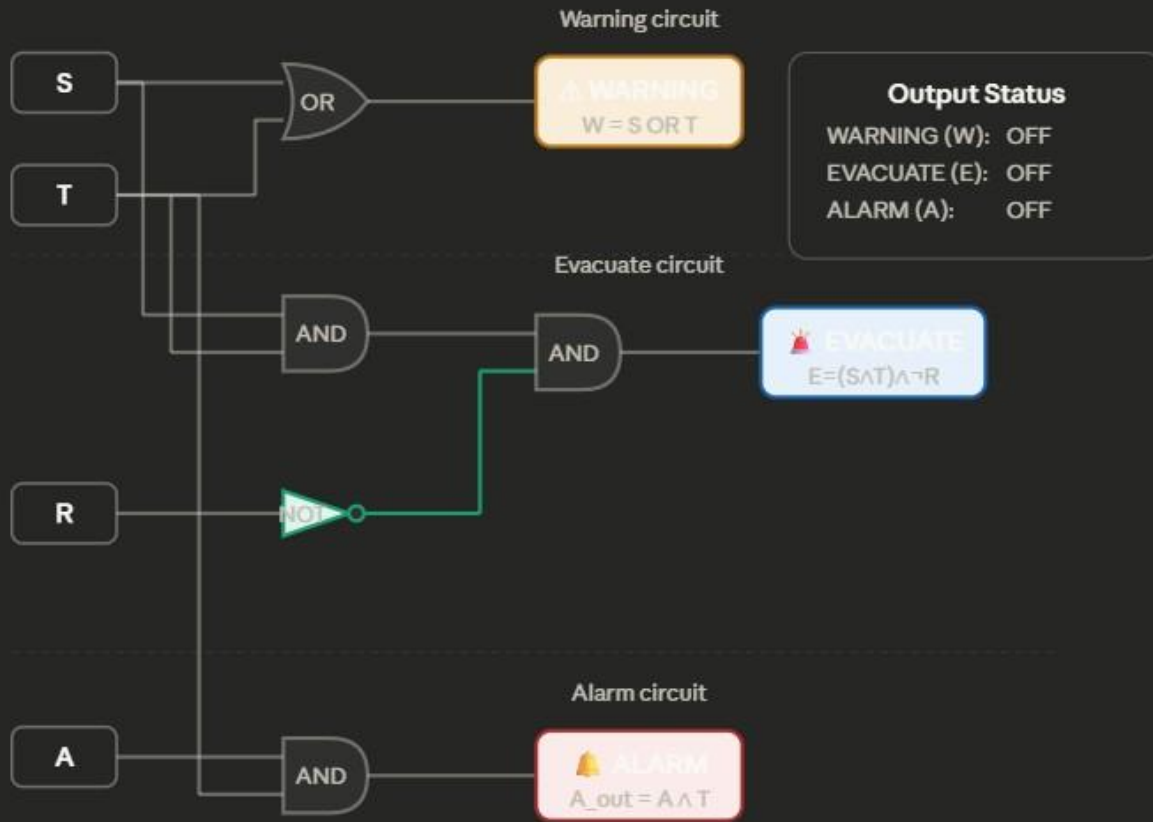
1. S= Smoke sensor
2. T=Temperature sensor
3. R=Reset Buton
4. M=Manual Test Switch

OUTPUTS:

1. W = Warning (YellowLed/ Slow Buzzer)
2. A = Alarm (Red Led, Fast Buzzer)
3. E = Evacuate (Continuous Buzzer+Blinking Red)

LOGIC CONCEPT:

STATUS	GREEN	YELLOW	RED	BUZZER
Safe	ON	×	×	Silent
Warning	×	ON	×	Slow
Alarm	×	×	ON	Fast
Evacuate	×	×	Flashing	Continuous
Reset	ON	×	×	Silent



WARNING

ALARM

EVACUATE

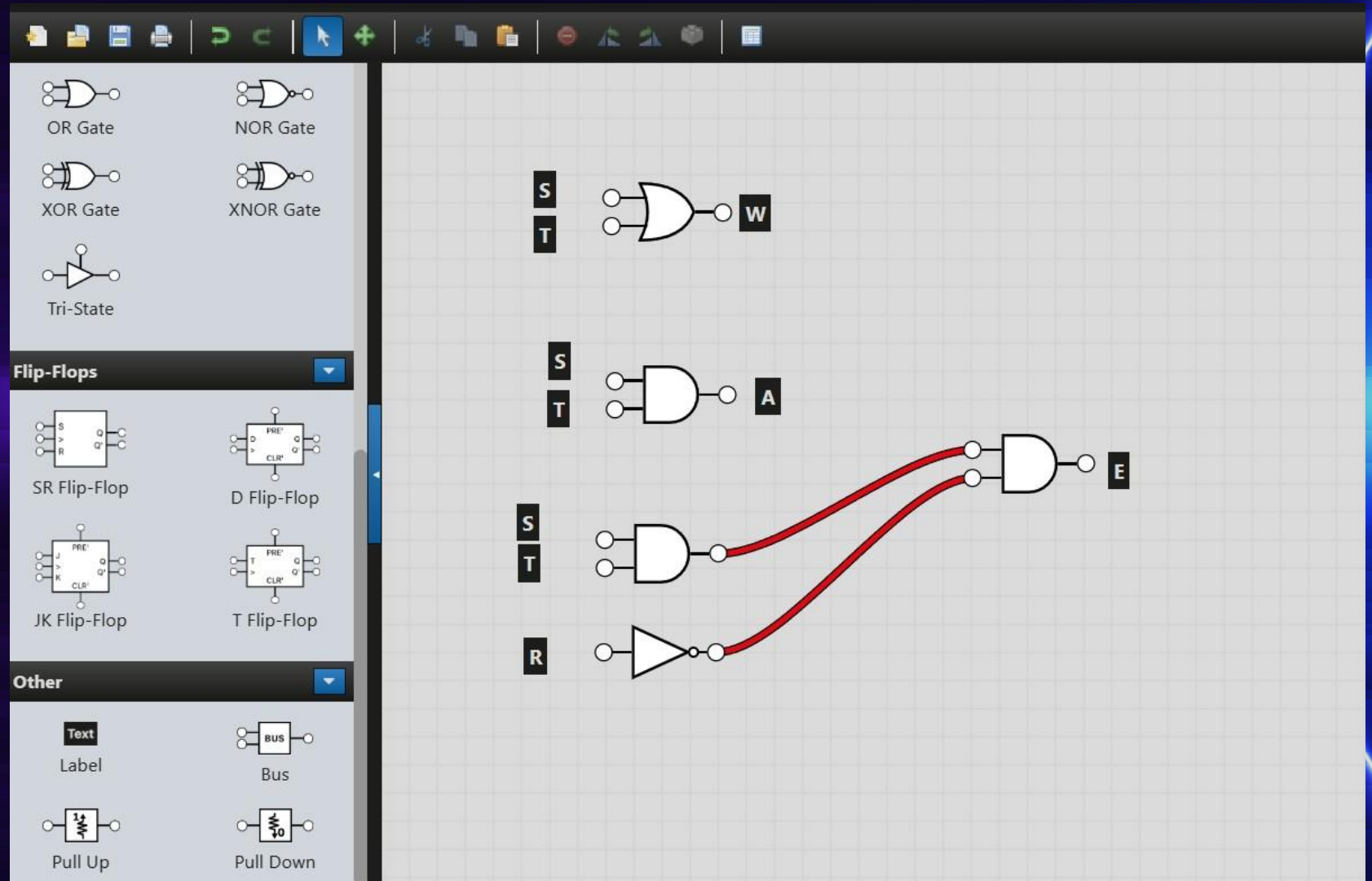
$W = S \text{ OR } T$

$A = A \text{ AND } T$

$E = (S \text{ AND } T)$
AND
NOT R

Logic Gate

LOGIC GATES DIAGRAM



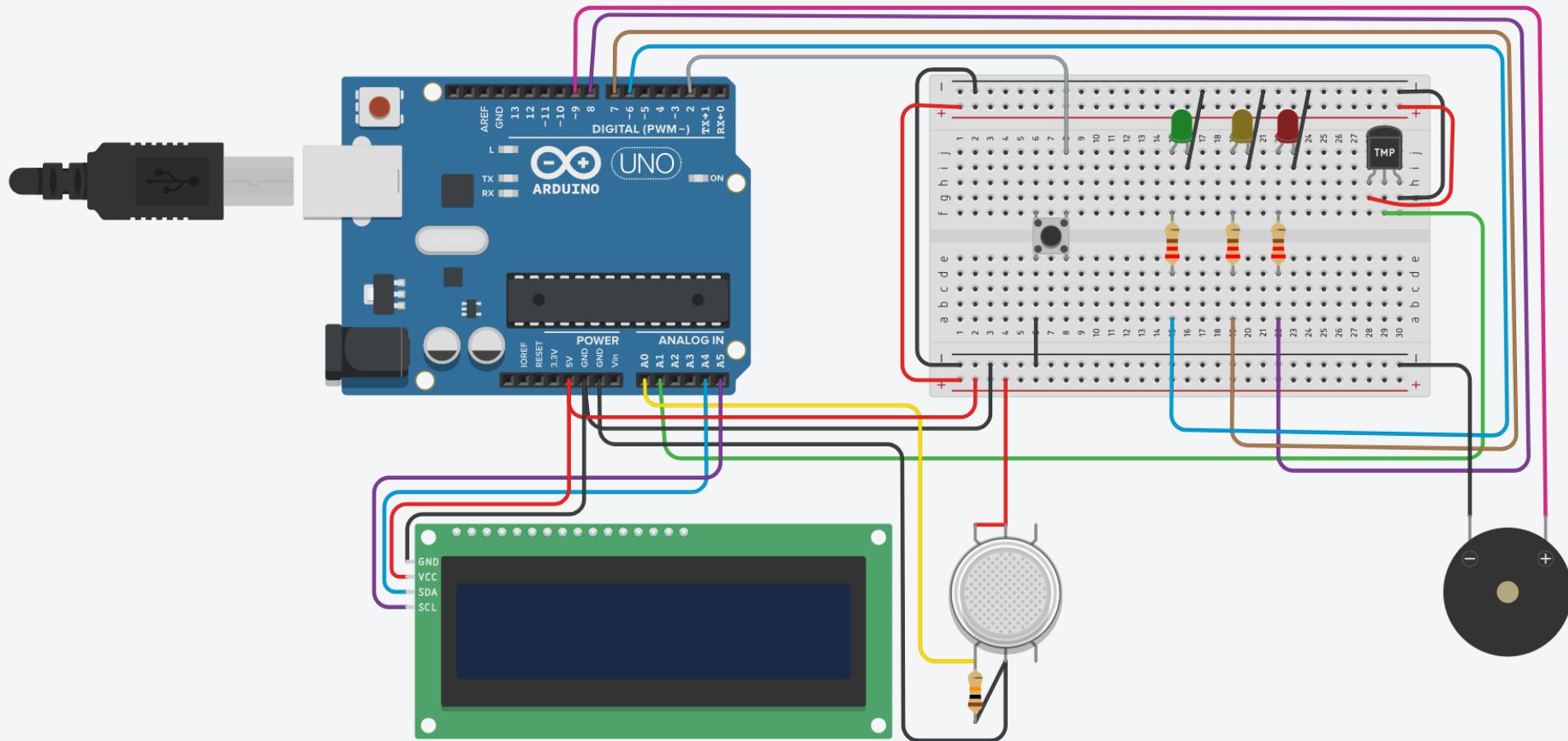
LOGIC.LY

TRUTH TABLE

W = OR A = AND E = (AND + NOT R)

S	T	R	W	A	E
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	1	0

TINKERCAD DESIGN



https://www.tinkercad.com/things/4fJy21QMP8D-amazing-hillar/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=hoFBjsEa2PsuOEFNwhZwsNTVHu7_IZqmhv1quBh3Kqw

TINKERCAD.COM

TINKERCAD C++

```
Text 1 (Arduino Uno R3)

1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 LiquidCrystal_I2C lcd(0x27, 16, 2); // Tinkercad ALWAYS uses 0x
5
6 const int gasPin = A0;
7 const int tempPin = A1;
8 const int resetPin = 2;
9 const int greenLED = 6;
10 const int yellowLED = 7;
11 const int redLED = 8;
12 const int buzzer = 9;
13
14 unsigned long alarmStart = 0;
15 bool timerOn = false;
16 String lastState = "";
17
18 // --- LCD Helper ---
19 void setLCD(const char* line1, const char* line2) {
20   lcd.clear();
21   lcd.setCursor(0, 0);
22   lcd.print(line1);
23   lcd.setCursor(0, 1);
24   lcd.print(line2);
25 }
26
27 // --- All outputs OFF ---
28 void alloff() {
29   digitalWrite(greenLED, LOW);
30   digitalWrite(yellowLED, LOW);
31   digitalWrite(redLED, LOW);
32   digitalWrite(buzzer, LOW);
33 }
34
35 void setup() {
36   pinMode(greenLED, OUTPUT);
37   pinMode(yellowLED, OUTPUT);
38   pinMode(redLED, OUTPUT);
39   pinMode(buzzer, OUTPUT);
40   pinMode(resetPin, INPUT_PULLUP);
41
42   Serial.begin(9600);
43
44 }
```

Serial Monitor

```
SAFE
Gas: 487 Temp: 24.78
RESET
Gas: 487 Temp: 24.78
SAFE
Gas: 487 Temp: 24.78
SAFE
```

```
Text 1 (Arduino Uno R3)

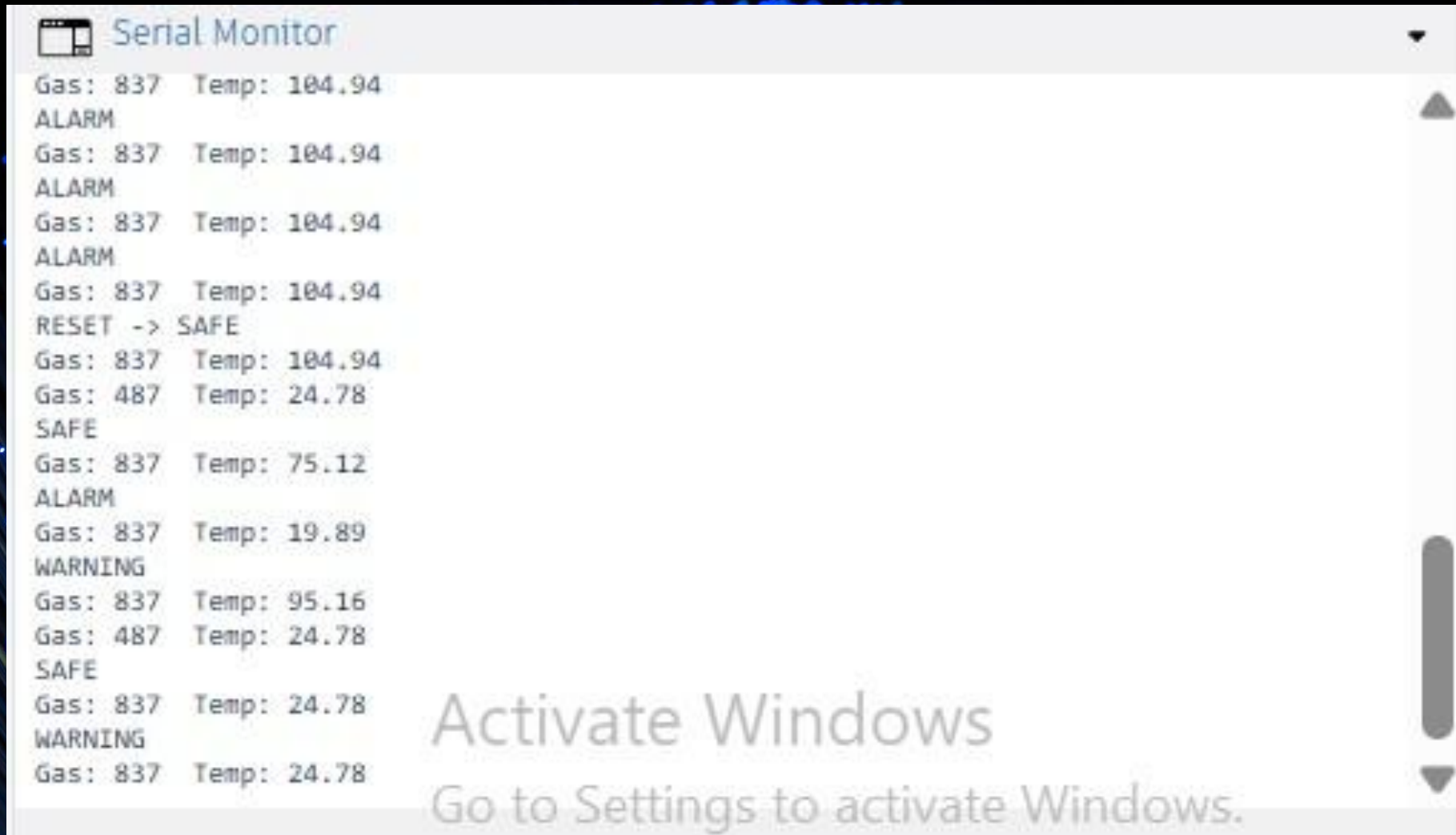
42 Serial.begin(9600);
43
44 // --- I2C + LCD INIT ---
45 Wire.begin();
46 delay(100); // Give LCD time to power up
47 lcd.init();
48 delay(50);
49 lcd.backlight();
50 delay(50);
51
52 setLCD(" Fire Detector ", " Starting... ");
53 delay(2000);
54
55 alloff();
56 digitalWrite(greenLED, HIGH);
57 setLCD(" SAFE ", " Monitoring... ");
58 lastState = "SAFE";
59 }
60
61 void loop() {
62   int gasVal = analogRead(gasPin);
63   float voltage = analogRead(tempPin) * (5.0 / 1023.0);
64   float tempC = (voltage - 0.5) * 100.0;
65   bool resetPressed = (digitalRead(resetPin) == LOW);
66
67   Serial.print("Gas: "); Serial.print(gasVal);
68   Serial.print(" Temp: "); Serial.println(tempC);
69
70   // --- RESET ---
71   if (resetPressed) {
72     alloff();
73     timerOn = false;
74     alarmStart = 0;
75     digitalWrite(greenLED, HIGH);
76     if (lastState != "RESET") {
77       setLCD(" RESET ", " System SAFE ");
78       lastState = "RESET";
79     }
80     Serial.println("RESET");
81     delay(500);
82     return;
83   }
84
85   // --- FIRE / EVACUATE ---
86   if (gasVal > 800 && tempC > 60) {
87     if (!timerOn) {
88       timerOn = true;
89       alarmStart = millis();
90     }
91
92     if (millis() - alarmStart > 10000) {
93       alloff();
94       if (lastState != "EVACUATE") {
95         setLCD("!!! FIRE !!! ", "EVACUATE NOW!! ");
96         lastState = "EVACUATE";
97         Serial.println("EVACUATE");
98       }
99       digitalWrite(buzzer, HIGH);
100      digitalWrite(redLED, HIGH);
101      delay(200);
102      digitalWrite(redLED, LOW);
103      delay(200);
104      return;
105    }
106  }
```

```
107   alloff();
108   if (lastState != "ALARM") {
109     setLCD(" !! ALARM !! ", " Temp+Gas High ");
110     lastState = "ALARM";
111     Serial.println("ALARM");
112   }
113   digitalWrite(redLED, HIGH);
114   digitalWrite(buzzer, HIGH);
115   delay(120);
116   digitalWrite(buzzer, LOW);
117   delay(120);
118   return;
119 }
120
121 timerOn = false;
122 alarmStart = 0;
123
124 // --- WARNING ---
125 if (gasVal > 700 || tempC > 35) {
126   alloff();
127   if (lastState != "WARNING") {
128     setLCD(" !! WARNING !! ", " Check Area ");
129     lastState = "WARNING";
130     Serial.println("WARNING");
131   }
132   digitalWrite(yellowLED, HIGH);
133   digitalWrite(buzzer, HIGH);
134   delay(600);
135   digitalWrite(buzzer, LOW);
136   delay(600);
137   return;
138 }
139
140 // --- SAFE ---
141 alloff();
142 digitalWrite(greenLED, HIGH);
143
144 if (lastState != "SAFE") {
145   setLCD(" SAFE : ) ", "Temp: ");
146   lastState = "SAFE";
147 }
148 // Live temp update on line 2
149 lcd.setCursor(5, 1);
150 lcd.print(tempC, 1);
151 lcd.print((char)223); // ° symbol
152 lcd.print("C ");
153
154 Serial.println("SAFE");
155 delay(500);
156 }
```

Serial Monitor

```
SAFE
Gas: 487 Temp: 24.78
RESET
Gas: 487 Temp: 24.78
SAFE
Gas: 487 Temp: 24.78
SAFE
```

TINKERCAD SERIAL MONITOR VIEW



TECHNICAL DEMO

Demo Platform: Tinkercad Circuits (online simulation)

What the demo shows:

- ✓ Normal state: Sensors reading low values → No alarm, LED off, buzzer silent
- ✓ Trigger state: Smoke level exceeds threshold → Buzzer activates, Red LED turns ON
- ✓ Code behaviour: Arduino reads analog input from MQ-2 → if value > 300, set buzzer and LED to HIGH

Component roles in demo:

MQ-2: Provides analog voltage signal proportional to smoke concentration

Arduino Uno: Processes signal and executes alarm logic

Buzzer: Produces audible alarm output

Red LED: Provides visible alert confirmation

Visual tip: Take a clear screenshot of your Tinkercad circuit in both OFF and ON states – place them side by side on the slide

RESULTS & EVALUATION

Results:

- ✓ System successfully detects simulated smoke input above threshold
- ✓ Buzzer and LED respond instantly and correctly
- ✓ Arduino logic matches the Boolean expression: $A = S + T$
- ✓ Tinkercad demo validates the design works as intended

Limitations:

- ⚠ Simulation only — real-world sensor calibration may vary
- ⚠ No remote alert (no WiFi/GSM module in current prototype)
- ⚠ MQ-2 can produce false positives (e.g. cooking fumes)

Next Steps:

Add ESP8266 WiFi module for mobile push notifications
Integrate with smart home platforms (Google Home, Alexa)
Add fire suppression trigger relay

SDG Contribution:

- **SDG 3:** Reduces risk of fire-related injury and death
- **SDG 11:** Contributes to building smarter, safer residential communities

REFERENCES:

Journal / Research Papers

1. Kunver Arif Ali et al., "Fire Alarm Systems." International Journal for Research in Applied Science and Engineering Technology, 2023.
2. Promise Elechi, "Design and Implementation of an Improved Smoke Detection System." European Journal of Electrical Engineering and Computer Science, 2017.
3. Mohamad Syafiq Mohamad Adenan et al., "Smart Smoke Detector." International Journal of Recent Technology and Applied Science, 2021.
4. Sreejith S. P., Kuldeep Baban Vayadande, "Advanced Fire Monitoring System." International Journal of Trend in Scientific Research and Development, 2020.
5. Robert S. Allison et al., "Sensors for Fire and Smoke Monitoring." Sensors Journal, 2021

I hope you understand
everything clearly.

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Thank you